# ENGINEERING PHYSICS CALCULATOR

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# CASIO SCIENTIFIC CALCULATOR

- A mini-project developed using C language.
- Implements Scientific , matrix , and equation-solving features.
- Also implements arithmetic operations.

# PROJECT OBJECTIVE :

- Create a menu-driven calculator.
- Implement reusable functions.
- Support advanced mathematics.
- Simulate CASIO calculator behavior.


# KEY FEATURES :

- Arithmetic operations (any count)
- Scientific functions
  **ex: calculates values of sin 30 , cos45, etc..**
- Matrix calculations
  **ex: mat addition, subtraction, multiplication, inverse**
- Equations solvers
  **ex: solves ax^2+bx+c**
- Ans memory feature
  **ex: stores last result**

# Program structure

- Header files and libraries
- Global “Ans” variable
- Modular functions
- Infinite menu loop
- Switch-case logic


# "Ans" MEMORY

## FEATURES :

- Global variable → double Ans
- Stores last calculated results
- Accessible in all operations
- Mimics CASIO calculator memory

# Basic Arithmetic Operations :

- Addition
- Subtraction
- Multiplication
- Division
- Supports multiple inputs using loops
  (included the special case)

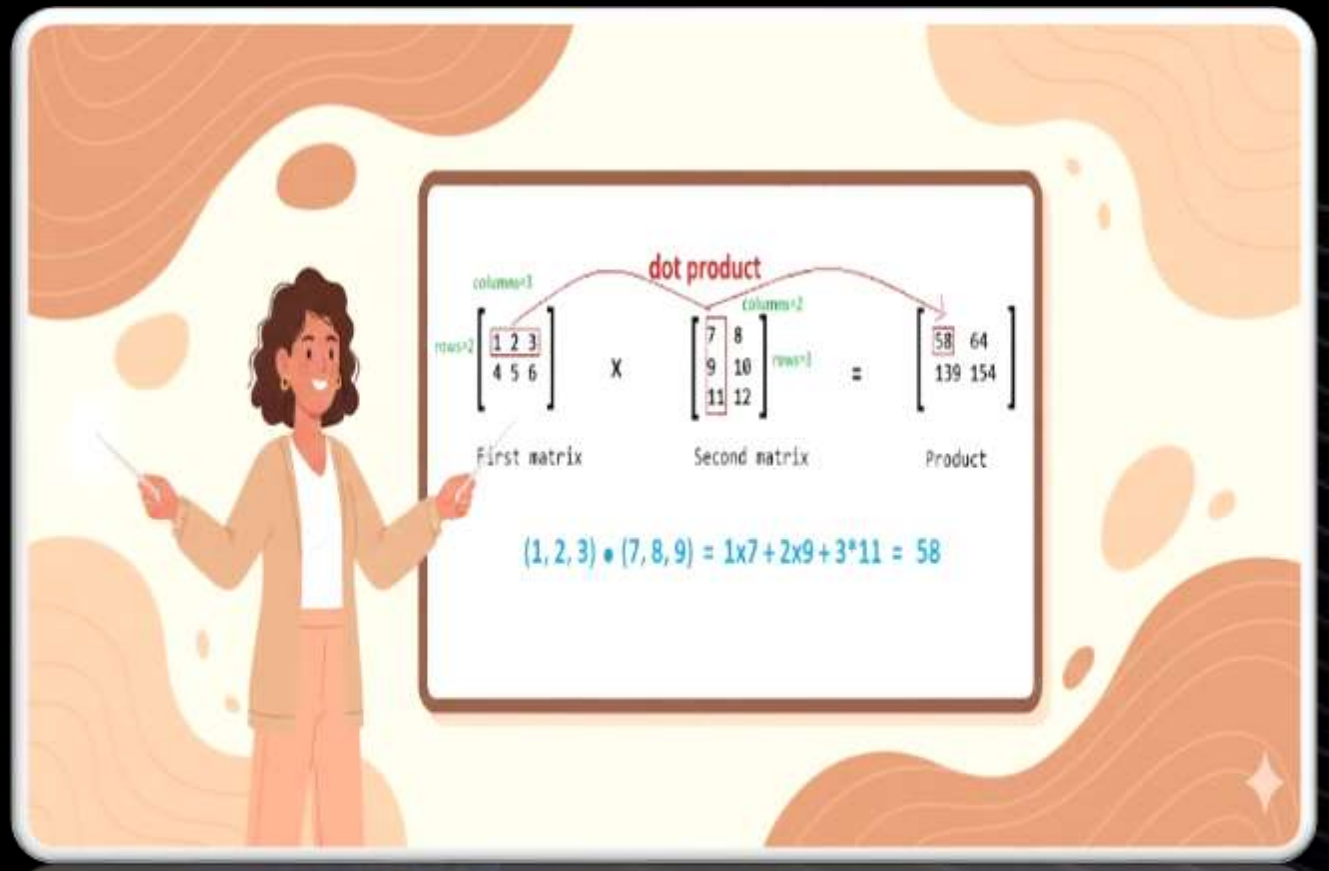
# COMBINATIONS AND STATISTICS


- Factorial of a number
- nPr (permutations) and nCr (combinations)
- Mean Calculation
- Max and Min operations
  **(finding maximum and minimum among the given numbers)**

# MATRIX OPERATIONS

- MATRIX ADDITION
  **(Must enter size of the matrices)**
- MATRIX MULTIPLICATION
  **(Based on 2 cases)**
- INVERSE OF THE MATRIX
- DIMENSTION VALIDATION

- **The code is declared globally**
  **i.e., Outside of the main function**
- **Call the respective functions when task is initialized**




# EQUATION SOLVERS:

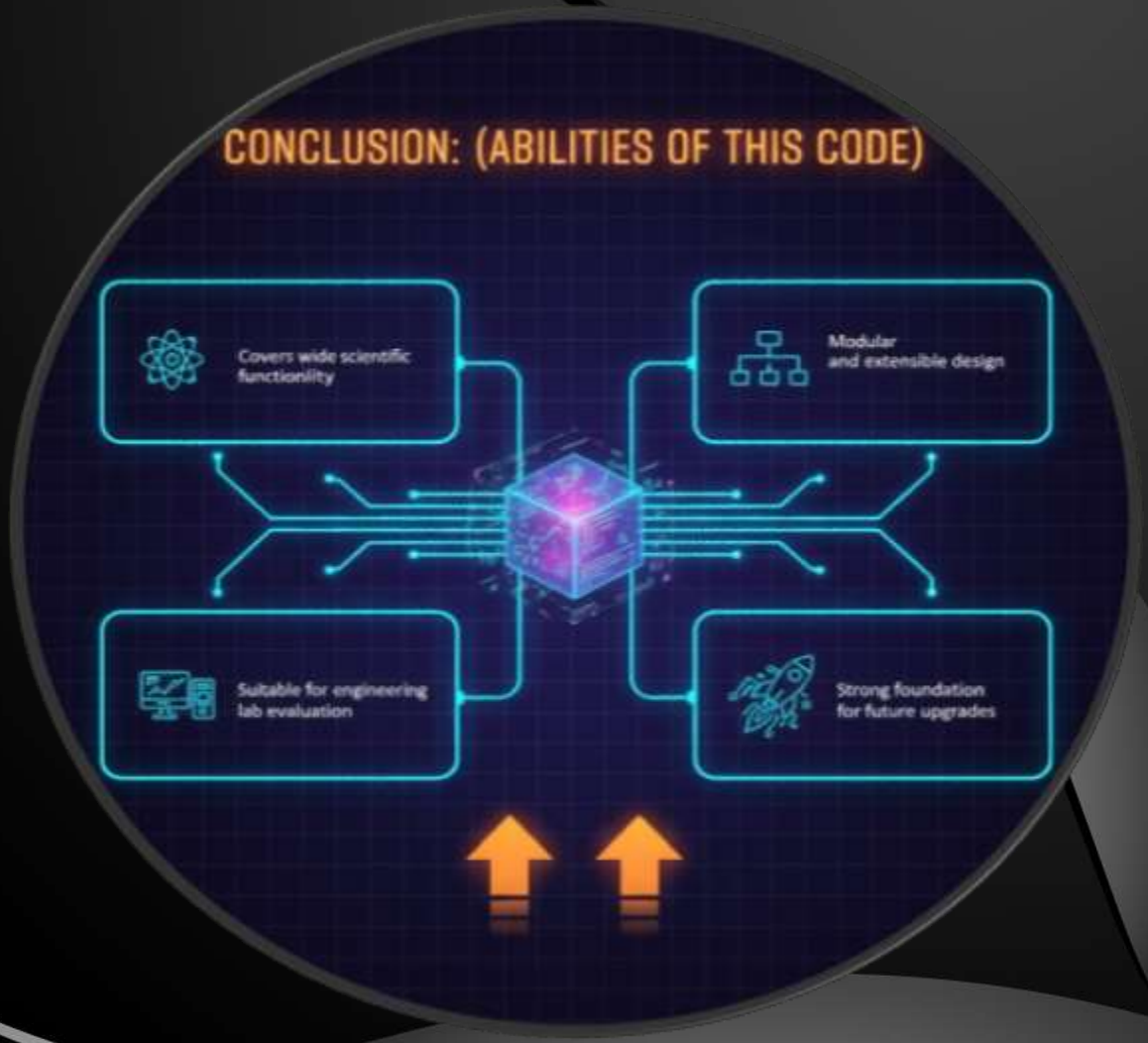
- Solves Quadratic equations using discriminants
  **(gives roots of the Q.E using 3 conditions)**
- Cubic equations using Newton-Raphson method
- Handles real and complex roots

- **The code is declared globally**
  **i.e., Outside of the main function**
- **Code is called by function present in the switch case**

# CONCLUSION:

## (Abilities of this code)

- Covers wide scientific functionality
- Modular and extensible design
- Suitable for engineering lab evaluation
- Strong foundation for future upgrades



# Code

```
#include <stdio.h>
#include <math.h>

double Ans = 0;   // Stores last answer

// ====================
MATRIX INVERSE (2x2 ONLY)
====================
void inverse2x2() {
  double a, b, c, d;
  printf("Enter elements of 2x2
matrix:\n");
  scanf("%lf %lf", Ca, Cb);
  scanf("%lf %lf", Cc, Cd);

  double det = a*d - b*c;

  if(det == 0) {
    printf("Matrix is singular! Inverse not
possible.\n");
    return;
  }

  printf("Inverse matrix:\n");
  printf("%.4lf %.4lf\n",  d/det, -b/det);
  printf("%.4lf %.4lf\n", -c/det,  a/det);
}

// ==================== QUADRATIC
ROOTS ====================
void quadratic() {
  double a,b,c;
  printf("Enter a, b, c: ");
  scanf("%lf %lf %lf", Ca, Cb, Cc);

  double D = b*b - 4*a*c;

  if(D > 0) {
    double r1 = (-b + sqrt(D)) / (2*a);
    double r2 = (-b - sqrt(D)) / (2*a);
    printf("Real and distinct
roots:\n%.4lf, %.4lf\n", r1, r2);
    Ans = r1;
  }
  else if(D == 0) {
    double r = -b / (2*a);
    printf("Real and equal roots:
%.4lf\n", r);
    Ans = r;
  }
  else {
    double real = -b / (2*a);
    double imag = sqrt(-D) / (2*a);
    printf("Complex roots:\n%.4lf +
%.4lfi\n%.4lf - %.4lfi\n", real, imag, real,
imag);
    Ans = real;
  }
}

// ==================== CUBIC
ROOTS (NUMERICAL NEWTON
METHOD) ====================
double cubic_f(double a,double
b,double c,double d,double x){
  return a*x*x*x + b*x*x + c*x + d;
}

double cubic_df(double a,double
b,double c,double x){
  return 3*a*x*x + 2*b*x + c;
}

void cubic() {
  double a,b,c,d;
  printf("Enter coefficients a b c d: ");
  scanf("%lf %lf %lf %lf",Ca,Cb,Cc,Cd);

  double x = 1;
  for(int i=0;i<10000;i++){
    double fx = cubic_f(a,b,c,d,x);
    double fdx = cubic_df(a,b,c,x);
    x = x - fx/fdx;
    Ans = x;
  }

  printf("One real root = %.6lf\n", x);
  printf("All the other remaining roots
are complex roots....\n");
}
```

```
// ==================== MATRIX ADD ====================
void matrix_add() {
  int r,c;
  printf("Enter rows and cols: ");
  scanf("%d %d",Cr,Cc);

  double A[10][10], B[10][10];

  printf("Enter Matrix A:\n");
  for(int i=0;i<r;i++)
    for(int j=0;j<c;j++)
      scanf("%lf",CA[i][j]);

  printf("Enter Matrix B:\n");
  for(int i=0;i<r;i++)
    for(int j=0;j<c;j++)
      scanf("%lf",CB[i][j]);

  printf("Sum:\n");
  for(int i=0;i<r;i++){
    for(int j=0;j<c;j++){
      printf("%.2lf ", A[i][j] + B[i][j]);
    }
    printf("\n");
  }
}
```

```
// ==================== MATRIX MULTIPLY ====================
void matrix_multiply() {
  int r1,c1,r2,c2;
  printf("Enter rows C cols of A: ");
  scanf("%d %d",Cr1,Cc1);
  printf("Enter rows C cols of B: ");
  scanf("%d %d",Cr2,Cc2);

  if(c1 != r2){
    printf("Invalid! Columns of A must equal rows of B.\n");
    return;
  }

  double A[10][10], B[10][10], C[10][10];
  printf("Enter A:\n");
  for(int i=0;i<r1;i++)
    for(int j=0;j<c1;j++)
      scanf("%lf",CA[i][j]);

  printf("Enter B:\n");
  for(int i=0;i<r2;i++)
    for(int j=0;j<c2;j++)
      scanf("%lf",CB[i][j]);

  for(int i=0;i<r1;i++){
    for(int j=0;j<c2;j++){
      C[i][j] = 0;
      for(int k=0;k<c1;k++)
        C[i][j] += A[i][k] * B[k][j];
    }
  }

  printf("Product:\n");
  for(int i=0;i<r1;i++){
    for(int j=0;j<c2;j++)
      printf("%.2lf ",C[i][j]);
    printf("\n");
  } }

long long factorial(int n) {
  long long fact = 1;
  for(int i = 1; i <= n; i++)
    fact *= i;
  return fact;
}

int main() {
  int choice;

  while (1) {
    printf("\n========== CASIO SCIENTIFIC CALCULATOR ==========\n");
    printf("Last Ans = %.4lf\n", Ans);
    printf("1. Addition (any number)\n");
    printf("2. Subtraction\n");
```

```
    printf("3. Multiplication\n");
    printf("4. Division\n");
    printf("5. Power chain\n");
    printf("6. Square root\n");
    printf("7. Factorial\n");
    printf("8. log10\n");
    printf("9. ln\n");
    printf("10. Sin\n");
    printf("11. Cos\n");
    printf("12. Tan\n");
    printf("13. nPr\n");
    printf("14. nCr\n");
    printf("15. Mean\n");
    printf("16. Max\n");
    printf("17. Min\n");
    printf("18. Matrix Addition\n");
    printf("19. Matrix Multiplication\n");
    printf("20. Matrix Inverse 2x2\n");
    printf("21. Quadratic Roots\n");
    printf("22. Cubic Equation (1 real
root)\n");
    printf("23. Use Ans\n");
    printf("0. Exit\n");
    printf("Enter choice: ");
    scanf("%d", Cchoice);

    if (choice == 0) break;

    int n;
    double num, result;
```

```
    switch (choice) {

      case 23:
        printf("Ans = %.4lf\n", Ans);
        continue;

      case 1:
        printf("How many numbers? ");
        scanf("%d", Cn);
        result = 0;
        for(int i=1;i<=n;i++){
          scanf("%lf",Cnum);
          result += num;
        }
        Ans = result;
        printf("Sum = %.4lf\n", result);
        break;

      case 2:
        printf("How many numbers? ");
        scanf("%d", Cn);
        scanf("%lf", Cresult);
        for(int i=2;i<=n;i++){
          scanf("%lf",Cnum);
          result -= num;
        }
        Ans = result;
        printf("Difference = %.4lf\n",
result);
        break;

      case 3:
        printf("How many numbers? ");
        scanf("%d", Cn);
        result = 1;
        for(int i=1;i<=n;i++){
          scanf("%lf",Cnum);
          result *= num;
        }
        Ans = result;
        printf("Product = %.4lf\n",
result);
        break;

      case 4:
        printf("How many numbers? ");
        scanf("%d", Cn);
        scanf("%lf",Cresult);
        for(int i=2;i<=n;i++){
          scanf("%lf",Cnum);
          if(num==0){
            printf("Error! Division by
zero.\n");
            break;
          }
          result /= num;
        }
        Ans = result;
```

```
        printf("Quotient = %.4lf\n",
result);
        break;

      case 5: {
        int count;
        printf("How many numbers? ");
        scanf("%d", Ccount);

        double arr[20];
        for(int i=0;i<count;i++)
          scanf("%lf",Carr[i]);

        double p = arr[0];
        for(int i=1;i<count;i++)
          p = pow(p, arr[i]);

        Ans = p;
        printf("Result = %.4lf\n", p);
        break;
      }

      case 6:
        printf("Enter number: ");
        scanf("%lf",Cnum);
        Ans = sqrt(num);
        printf("sqrt  = %.4lf\n", Ans);
        break;

      case 7:
        printf("Enter number: ");
        scanf("%lf",Cnum);
        Ans = factorial(num);
        printf("Factorial = %.0lf\n", Ans);
        break;

      case 8:
        printf("Enter number: ");
        scanf("%lf",Cnum);
        Ans = log10(num);
        printf("log10 = %.4lf\n", Ans);
        break;

      case 9:
        printf("Enter number: ");
        scanf("%lf",Cnum);
        Ans = log(num);
        printf("ln = %.4lf\n", Ans);
        break;

      case 10:
        printf("Enter degrees: ");
        scanf("%lf",Cnum);
        Ans = sin(num * M_PI/180);
        printf("sin = %.4lf\n", Ans);
        break;

      case 11:
        printf("Enter degrees: ");
        scanf("%lf",Cnum);
        Ans = cos(num * M_PI/180);
        printf("cos = %.4lf\n", Ans);
        break;

      case 12:
        printf("Enter degrees: ");
        scanf("%lf",Cnum);
        Ans = tan(num * M_PI/180);
        printf("tan = %.4lf\n", Ans);
        break;

      case 13: {
        int n, r;
        printf("Enter n, r: ");
        scanf("%d %d", Cn, Cr);
        Ans = factorial(n) / factorial(n-
        r); printf("nPr = %.0lf\n", Ans);
        break;
      }

      case 14: {
        int n, r;
        printf("Enter n, r: ");
        scanf("%d %d", Cn, Cr);
        Ans = factorial(n) /
(factorial(r)*factorial(n-r));
        printf("nCr = %.0lf\n", Ans);
        break;
      }
```

```
case 15:
  printf("How many numbers? ");

   scanf("%d",Cn);
   result = 0;
   for(int i=1;i<=n;i++){
      scanf("%lf",Cnum);
      result += num;
   }
   Ans = result/n;
   printf("Mean = %.4lf\n", Ans);
   break;

case 16:
  printf("How many numbers? ");
   scanf("%d",Cn);
   scanf("%lf",Cresult);
   for(int i=2;i<=n;i++){
      scanf("%lf",Cnum);
     if(num > result) result = num;
   }
   Ans = result;
   printf("Max = %.4lf\n", Ans);
   break;

case 17:
  printf("How many numbers? ");
   scanf("%d",Cn);
   scanf("%lf",Cresult);
   for(int i=2;i<=n;i++){
      scanf("%lf",Cnum);
      if(num < result) result = num;
   }

   Ans = result;
   printf("Min = %.4lf\n", Ans);
   break;

case 18:
   matrix_add();
   break;

case 19:
   matrix_multiply();
   break;

case 20:
   inverse2x2();
   break;

case 21:
   quadratic();
   break;

case 22:
   cubic();
   break;

default:
   printf("Invalid choice!\n");
  }
}
```

```
   return
   0;
}
```

# Thank You